Product Requirements Document

Cake Slice Factory — Sales, Inventory & Expense Management

Field	Value
Product	Cake Slice Factory
Version	1.0 (Draft for approval)
Status	Ready for review
Document owner	Business owner (product sponsor)
Prepared by	Engineering
Last updated	2026-10-21
Stack	HTML, CSS, JavaScript, Supabase (Postgres + Auth + Storage + Realtime)

1. Overview

1.1 Problem statement

Cake Slice Factory sells cakes through walk-in customers and via Instagram and WhatsApp, and bakes from raw ingredients. Today orders, ingredient stock, production and expenses are tracked by hand (notebooks, WhatsApp chats, memory). The consequences are:

No reliable picture of what is in stock (ingredients or finished cakes).

Orders taken on Instagram/WhatsApp are easy to lose or forget, and delivery dates slip.

Prices are guessed because true cost per cake (from the recipe) is not known.

Money leaks: expenses are unrecorded, so profit is unknown.

No accountability — anyone can change a price or delete a record with no trace.

Repeating customers are not recognised, and top locations are unknown, so no marketing decisions.

1.2 Product vision

One simple, mobile-friendly web app that runs the whole operation: take an order from any channel, track the ingredients and finished cakes it consumes, record every naira in and out, and print a professional quote/invoice/receipt — all with an accurate profit picture for the owner.

1.3 Objectives (12 months)

#	Objective	Target
O1	Every sale is recorded in the system	≥ 95% of orders
O2	No order forgotten	0 missed fulfilment dates
O3	Ingredient stock is knowable at any time	100% of stock movements logged
O4	True cost & margin per product	Known for all active products
O5	Know monthly profit	Report available within 1 day of month end
O6	Accountability for edits	100% of money-affecting changes audited

1.4 Success metrics (KPIs)

Adoption: % of orders entered per week; active staff users per day.

Data quality: % of orders with a linked customer; % of products with a recipe.

Business value: gross margin % trend; stock-out incidents per month; unrecorded-expense count (target 0).

Speed: median time to record a walk-in order (< 45 seconds).

Reliability: order data loss incidents (target 0).

2. Personas & Roles

Persona	Description	Primary goals	Device
Owner / Sponsor	Runs the business, prices products, decides.	Profit visibility, control, accountability.	Phone + laptop
Front-desk staff	Takes orders at the counter and via Instagram/WhatsApp.	Record an order fast; issue a receipt; answer "do we have it?"	Phone (counter)

Baker / Production staff	Bakes, records output, consumes ingredients.	Log a batch, know what's in the store.	Phone/tablet
Accountant (Phase 2)	Periodic review of expenses and payables.	Clean expense data, exportable.	Laptop

2.1 Role & permission matrix

Capability	Owner	Staff
View dashboard & reports	☑	☑
Create/edit/delete products, ingredients, vendors, recipes	☑	✗ (read-only)
Set selling prices & discounts	☑	⚠ discount capped, logged
Create/edit orders, customers, payments	☑	☑
Complete / cancel an order	☑	⚠ configurable
Void or delete a completed order	☑	✗
Record production batches	☑	☑
Record items received / purchases	☑	☑
Record expenses	☑	☑
Run stock take	☑	☑
Manage users & roles	☑	✗
View audit log	☑	✗
Export data / backup	☑	✗

Permissions are enforced in the database with Row Level Security, not just in the UI.

3. Scope

3.1 In scope

Sales (walk-in, Instagram, WhatsApp, phone), customers, orders (ready-made and custom cakes), a quotes workspace, delivery vs pick-up, a month calendar and dispatch board, issued quotes/invoices/receipts, a payment ledger with balances, finished-goods inventory, ingredient inventory, recipes, vendor records, items received, production batches, stock ledger, expenses, reporting and analytics, multi-format exports, users and permissions.

3.2 Out of scope (v1)

Payroll calculation, full double-entry accounting, e-commerce storefront, native mobile apps, multi-currency, multi-branch, automated marketing campaigns, barcode hardware.

3.3 Future scope (v2+)

Multi-branch, customer self-service order form, loyalty/promotions, demand forecasting, vendor price comparison, barcode scanning, offline-first sync with conflict resolution.

4. Features Added Beyond the Original Brief

The owner listed Sales, Inventory and Expenses. The following features were not requested but are necessary for the system to be safe, trustworthy and actually usable in a real cake business. Each is prioritised with MoSCoW (Must, Should, Could, Won't-yet).

4.1 Critical — the system is unsafe without these

#	Feature	Why it is necessary	Priority
A1	Users, roles & permissions	Money-handling systems need accountability. Anyone editing prices/stock must be identified.	M
A2	Audit / activity log	"Who changed this price? Who deleted that order?" must be answerable. Immutable history of money-affecting changes.	M
A3	Refunds, cancellations & stock reversal	Stock is deducted when an order completes. Without a reversal path, cancelling an order permanently corrupts stock.	M
A4	Balances, debtors & part payments	Real customers pay a deposit then balance. Without this, "who owes me money?" is unanswerable.	M
A5	Discounts & price overrides (logged)	Discounting is normal; undisciplined discounting destroys margin. Must be captured and attributed.	M
A6	Waste, spoilage & returns	Cakes burn, ingredients expire, items are returned. Unrecorded waste silently breaks stock accuracy.	M

A7	Data export & backup	Business continuity. The owner must be able to extract their own data as CSV, Excel, Word, PDF or print, and restore from a full JSON snapshot.	M
A8	Offline / poor-network resilience	Nigerian mobile data drops mid-order. The app must not silently lose a sale.	M
A9	Soft delete + restore (trash)	Accidental deletes must be recoverable; history must never break.	M
A10	Input validation & clear error handling	Bad data (negative stock, missing phone) destroys reports.	M
A11	Mobile-first responsive UI	Orders are taken on a phone at the counter. Desktop-only fails.	M
A12	Tax (VAT) handling & rounding rules	Invoices must be legally correct and totals must reconcile to the kobo.	M
A13	Branded document printing	Quote/invoice/receipt must carry business name, address, phone and logo to be professional.	M
A14	Low-stock alerts & reorder points	The whole point of inventory — know before you run out.	M
A15	Search, filter, sort, pagination	With hundreds of products/orders, a plain list is unusable.	M

4.2 Important — needed for real day-to-day operation

#	Feature	Why it matters	Priority
B1	Customer CRM & purchase history	Repeat customers are the core of a cake business; enables "last order again".	S
B2	WhatsApp deep-link to send quote/invoice/receipt	Matches how the business actually communicates; removes manual re-typing.	S
B3	Production calendar & order scheduling	Prevents double-booking the oven and missed delivery dates.	S
B4	Unit conversion (bag ↔ kg ↔ g)	Ingredients are bought in 50 kg bags but recipes use grams. Without this, recipes are wrong.	S

B5	Recipe costing & margin analysis	Shows true cost per cake and profit per product → correct pricing.	S
B6	Purchase orders & vendor payables	Order ingredients ahead; track what you owe vendors.	S
B7	Stock take / physical count & variance	The only way to detect and correct drift between system and reality.	S
B8	Batch & expiry tracking	Perishable ingredients (dairy, eggs) must be used oldest-first.	S
B9	Delivery/rider assignment & delivery status	Deliveries fail silently without ownership of the trip.	S
B10	Attachments (cake reference photos, expense receipts)	Evidence for custom orders and expense claims.	S
B11	Price history	Explains price changes; supports margin trend analysis.	S
B12	In-app notifications (low stock, due orders)	Turns the system from passive record to active assistant.	S
B13	Daily sales close / cash-up summary	End-of-day reconciliation between cash received and orders recorded.	S
B14	Profit & Loss and cash-flow reports	The owner's core question: "did I make money?"	S
B15	Void / override controls	Fraud prevention: restricts who can void and requires a reason.	S
B16	Stock movement reason codes	Every adjustment must say why, or the ledger is useless.	S
B17	Password reset & session management	Users must self-serve; sessions must expire safely.	S
B18	Onboarding & guided setup	A blank system is never adopted; guided first-run matters.	S

4.3 Nice to have

#	Feature	Priority

C1	Multi-branch / location support	C
C2	Barcode / QR scanning for products	C
C3	Loyalty points & promotions engine	C
C4	Vendor price comparison & best-vendor suggestions	C
C5	Demand forecasting from historical sales	C
C6	Dark mode	C
C7	Multi-currency	C
C8	Public customer order form (self-service)	C
C9	Two-factor authentication for owner accounts	C
C10	Automated off-site backup schedule	C

5. Functional Requirements

Each requirement has a stable ID (M<module>-FR-<n>), a priority, and acceptance criteria (AC).

M1 — Identity & Access

ID	Requirement	Pri	Acceptance criteria
M1-FR-00	First-run owner setup	M	The first signed-in account claims owner once from the sign-in page; a second claim is refused by the database
M1-FR-01	Users sign up / sign in with email + password	M	Valid credentials issue a session; invalid show an inline error
M1-FR-02	Password reset by email	M	User receives link and can set a new password
M1-FR-03	Owner can invite/create staff accounts and set role	M	New account defaults to staff; owner can set owner/staff

M1-FR-04	Owner can deactivate a staff member	S	Deactivated user cannot sign in; their history is retained
M1-FR-05	Sessions persist across reloads and auto-refresh	M	Reload keeps user signed in; expired session redirects to login
M1-FR-06	All data access is row-level filtered by role	M	Staff cannot write to products/ingredients/vendors/recipes via API

M2 — Product Catalog (finished cakes)

ID	Requirement	Pri	Acceptance criteria
M2-FR-01	Create a product with name, SKU, category, flavour, size, unit, price, cost, stock, low-stock level, image, status	M	Product appears in list and is searchable immediately
M2-FR-02	Distinguish ready-made vs custom (made-to-order) products	M	Custom products do not require tracked stock
M2-FR-03	Edit any product field	M	Changes persist; an audit entry is written; price change recorded in price history
M2-FR-04	Soft-delete (archive) a product; restore from trash	M	Archived product is hidden from new orders but retained in history
M2-FR-05	Duplicate a product (fast variant creation)	C	New row is created with " (copy)" suffix
M2-FR-06	Search by name/SKU; filter by category, kind, status, stock level	M	Filters combine correctly and return within 1s
M2-FR-07	Show computed margin % per product	S	margin = (price − cost) / price

M3 — Ingredients & Recipes

ID	Requirement	Pri	Acceptance criteria
M3-FR-01	Create/edit ingredients with unit, cost per unit, stock, low-stock level, preferred vendor	M	Saved and reflected in stock list

M3-FR-02	Define a recipe (bill of materials) per product	M	Recipe shows ingredients with quantities and rolled-up cost
M3-FR-03	Recipe cost rolls up into product cost	S	Product cost updates when ingredient costs change (recalc action)
M3-FR-04	Unit conversions per ingredient (e.g. 1 bag = 50 kg)	S	Purchase in bags; recipe consumes kg correctly
M3-FR-05	Batch/lot + expiry per received quantity	S	Expiry warnings surface ingredients near expiry
M3-FR-06	Low-stock and expiry alert list	M	Alerts appear on dashboard and are filterable

M4 — Vendors & Purchasing

ID	Requirement	Pri	Acceptance criteria
M4-FR-01	Create/edit vendor with contact, phone, email, address, supplies	M	Vendor selectable in purchases and expenses
M4-FR-02	Record items received (ingredient, qty, unit cost, date, vendor)	M	Ingredient stock increases and a ledger row is written
M4-FR-03	Update ingredient cost from latest purchase	S	cost_per_unit reflects most recent non-zero unit cost
M4-FR-04	Raise a purchase order to a vendor and mark received	S	Received PO creates purchase rows and stock movements
M4-FR-05	Vendor payables summary	S	Total owed per vendor = received POs − payments

M5 — Production

ID	Requirement	Pri	Acceptance criteria
M5-FR-01	Record a production batch (product, qty, date, note)	M	Batch number auto-generated
M5-FR-02	Consume recipe ingredients on batch creation	M	Ingredient stock decreases by recipe qty × batch qty; ledger rows written

M5-FR-03	Increase finished-cake stock on batch creation	M	Product stock increases by quantity produced
M5-FR-04	Warn when producing more than available ingredients	M	User is warned and must confirm an override (logged)
M5-FR-05	Production calendar showing batches and order due dates	S	Dated view of work outstanding
M5-FR-06	Record waste/spoilage against a batch	M	Waste logged with reason; stock adjusted

M6 — Sales & Orders

ID	Requirement	Pri	Acceptance criteria
M6-FR-01	Record customer name and phone on every order	M	Order cannot be saved without a customer name
M6-FR-02	Capture channel: walk-in, Instagram, WhatsApp, phone, other	M	Channel recorded and reportable
M6-FR-03	Add multiple line items (product, qty, unit price)	M	Line totals and order subtotal compute automatically
M6-FR-04	Create custom cake orders with free-form details and price	M	Custom details stored and shown on documents
M6-FR-05	Choose delivery or pick-up	M	Delivery requires an address; pick-up does not
M6-FR-06	Capture delivery/pick-up date and address	M	Date required; address required when delivery
M6-FR-07	Apply discounts (amount or %) with permission + reason	M	Discount stored, attributable, shown on documents
M6-FR-08	Order status pipeline: quote → confirmed → in_production → ready → out_for_delivery → completed / cancelled	M	Status changes are timestamped and attributed
M6-FR-09	Deduct finished-cake stock once when an order is completed	M	Stock decreases exactly once per order

M6-FR-10	Reverse stock when a completed order is cancelled or refunded	M	Stock returns; reversal ledger rows written; never double-reversed
M6-FR-11	Reorder a previous order in one click	C	New draft order copies prior line items
M6-FR-12	Search/filter orders by number, customer, status, channel, date	M	Results correct and fast
M6-FR-13	Assign a rider and track delivery status	S	Delivery has an owner and a status
M6-FR-14	Prevent duplicate order submission	M	Double-click/submit does not create two orders

M7 — Payments & Documents

ID	Requirement	Pri	Acceptance criteria
M7-FR-01	Record payments/deposits against an order	M	amount_paid and balance update automatically
M7-FR-02	Support cash, transfer, POS, other	M	Method recorded and reportable
M7-FR-03	Show outstanding balance and a debtors list	M	Orders with balance > 0 are listed with amount owed
M7-FR-04	Record a refund (negative payment)	M	Refund reduces amount_paid and appears in cash reports
M7-FR-05	Generate a quote, invoice and receipt per order	M	Each has an auto number (QUO-/INV-/RCT-)
M7-FR-06	Print / PDF exported documents with business branding	M	Business name, address, phone and logo present
M7-FR-07	Share a document via WhatsApp deep link	S	Opens WhatsApp with a pre-filled message
M7-FR-08	Documents are immutable snapshots	S	Re-printing shows the values at issue time

M7-FR-09	Payment ledger by method with refunds and outstanding balances	M	Every deposit, balance and refund is visible and reconciles to the order
M7-FR-10	Document register for issued quotes, invoices and receipts	M	Owner can find, reprint or export any issued document
M7-FR-11	Quotes workspace separate from confirmed orders	S	Open quotes can be reviewed without mixing them into live production
M7-FR-12	Month calendar and delivery dispatch board	S	Due work is grouped by date; orders move ready → out for delivery → completed

M8 — Customers & CRM

ID	Requirement	Pri	Acceptance criteria
M8-FR-01	Maintain customer records (name, phone, email, address, city, state, channel)	M	Customer is searchable and reusable across orders
M8-FR-02	Detect existing customer by phone and auto-fill	M	Typing a known phone populates the customer
M8-FR-03	View a customer's order history and lifetime spend	S	History and total shown on the customer page
M8-FR-04	City/state captured for location reporting	M	Orders can be grouped by location

M9 — Expenses

ID	Requirement	Pri	Acceptance criteria
M9-FR-01	Record ingredient purchases with the vendor	M	Purchase is recorded and linked to ingredient stock
M9-FR-02	Record other expenses with category (logistics, salaries, rent, utilities, marketing, equipment, packaging, other)	M	Categorised and reportable
M9-FR-03	Attach a receipt photo to an expense	S	Image stored in Supabase Storage and viewable
M9-FR-04	Optionally link an expense to an order (job costing)	S	Order profit reflects linked costs

M9-FR-05	Expense summary by month and category	M	Totals reconcile to the underlying rows

M10 — Reporting & Analytics

ID	Requirement	Pri	Acceptance criteria
M10-FR-01	Dashboard KPIs: today's orders/revenue, month revenue, total stock value, low-stock count, outstanding balance	M	Numbers match underlying data
M10-FR-02	Highest-selling product (by qty and by revenue)	M	Correct ranking for a selected period
M10-FR-03	Top customer by spend	M	Correct ranking for a selected period
M10-FR-04	Highest-revenue location (city/state)	M	Correct ranking for a selected period
M10-FR-05	Ingredients restocked the most, with quantity and amount	M	Correct ranking
M10-FR-06	Sales trend over time (day/week/month)	S	Chart renders for the period
M10-FR-07	Profit & Loss (revenue − COGS − expenses) by period	S	Totals reconcile to orders and expenses
M10-FR-08	Daily sales close / cash-up summary	S	Day's totals by payment method
M10-FR-09	Export any list or report to CSV, Excel, Word, PDF or print	M	Export chooser offers all five formats and each output opens cleanly

M11 — Platform & Administration

ID	Requirement	Pri	Acceptance criteria
M11-FR-01	Business settings: name, address, phone, currency, tax rate, logo, receipt footer	M	Settings appear on documents

M11-FR-02	Audit log of create/update/delete on money-affecting tables	M	Owner can filter by user, table, date
M11-FR-03	Trash / restore for soft-deleted records	M	Deleted item restorable by owner
M11-FR-04	Notifications: low stock, orders due today, overdue balances	S	Listed in-app with unread state
M11-FR-05	Onboarding checklist / guided live setup	S	A new install can be configured safely
M11-FR-06	Realtime updates when two users are online	S	Stock changes appear without reload

M12 — Data & Compliance

ID	Requirement	Pri	Acceptance criteria
M12-FR-01	Manual full export (all tables) as CSV/JSON	M	Owner can download everything
M12-FR-02	Soft delete everywhere; no hard deletes from the UI	M	No UI path can irreversibly delete money records
M12-FR-03	Customer data used only for orders; deletable on request	S	Customer anonymisation action available to owner
M12-FR-04	Referential integrity across all relationships	M	Deleting a vendor keeps history (set null)

6. Non-Functional Requirements

ID	Area	Requirement
NFR-01	Performance	First meaningful paint < 3 s on 3G; typical list/search response < 1 s for 10k records
NFR-02	Scale	Support ≥ 5,000 products, 100,000 orders, 500,000 stock movements without redesign

NFR-03	Availability	≥ 99.5% monthly (inherits Supabase SLA)
NFR-04	Security	TLS in transit; encryption at rest; RLS on every table; anon key only in client; no service-role key shipped
NFR-05	Privacy	Customer phone numbers accessible only to authenticated staff; not exposed publicly
NFR-06	Reliability	Order submission must not silently fail; failed writes are surfaced and retryable
NFR-07	Connectivity	A dropped connection shows a clear warning; writes require the live database and are never silently queued
NFR-08	Usability	Walk-in order recordable in < 45 s by a first-time user after a short walkthrough
NFR-09	Accessibility	Keyboard navigable, WCAG 2.1 AA contrast, labelled form fields
NFR-10	Compatibility	Latest Chrome, Safari, Edge, Firefox; Android Chrome and iOS Safari primary
NFR-11	Localisation	Currency ₦ (NGN), locale en-NG, DD/MM/YYYY dates; structured for future locales
NFR-12	Maintainability	No build step; ES modules; one module per domain; documented schema migrations
NFR-13	Observability	Client errors logged with context; audit log for business-critical changes
NFR-14	Backup/Restore	Database backups are controlled in Supabase; owner can download a full JSON snapshot; restore procedure documented
NFR-15	Time	All timestamps stored UTC, displayed in local time
NFR-16	Theme & navigation	Every page has a labelled desktop menu, a closable mobile drawer, and a persistent light/dark theme toggle

7. Business Rules

ID	Rule
BR-01	An order cannot be saved without a customer name and at least one line item or custom description.
BR-02	A delivery order requires a delivery address; a pick-up order does not.
BR-03	A delivery/pick-up date is required for any order that is not an open quote.
BR-04	Stock is deducted from finished cakes exactly once, when the order becomes completed.
BR-05	Cancelling or refunding a completed order reverses the stock deduction; reversals are idempotent.
BR-06	Custom (made-to-order) products do not hold stock and are never deducted.
BR-07	Creating a production batch consumes recipe ingredients and increases finished-cake stock atomically.
BR-08	Producing more than the available ingredients requires an explicit, logged override.
BR-09	Stock is allowed to go negative only with an owner override; otherwise the action is blocked with a warning.
BR-10	Only the owner can create, edit or delete products, ingredients, vendors and recipes.
BR-11	Only the owner can void or delete a completed order; staff must request it.
BR-12	Discounts above a configurable threshold require the owner role.
BR-13	Every stock change must carry a reason code.
BR-14	Order totals are always recalculated from line items: total = subtotal − discount + delivery_fee + tax.

BR-15	amount_paid is always the sum of payments (refunds are negative payments).
BR-16	Quote/Invoice/Receipt numbers are unique and sequential.
BR-17	Records are soft-deleted; history and reports must never show gaps from deletions.
BR-18	Ingredient cost per unit updates from the most recent purchase with a non-zero unit cost.
BR-19	A customer is matched by phone number (normalised) before creating a new record.
BR-20	All money is stored to 2 decimal places and rounded half-up.

8. Data Requirements (summary)

Primary entities and their purpose are detailed in ARCHITECTURE.md §4. Summary: profiles, settings, vendors, ingredients, products, recipes, customers, orders, order_items, payments, documents, production_batches, stock_movements, ingredient_purchases, expenses, plus the additions in supabase/002_core_additions.sql (audit_log, purchase_orders, purchase_order_items, stock_takes, stock_take_items, attachments, notifications, unit_conversions, price_history, order_status_history).

9. Release Plan

Release	Contents	Exit criteria
R0 — Foundation	Schema, RLS, auth, app shell, generic CRUD	Owner can sign in and manage products
R1 — Inventory	Products, ingredients, vendors, recipes, low-stock alerts	Stock can be maintained accurately
R2 — Sales	Customers, orders, custom cakes, payments, documents	A real order can be recorded and receipted end-to-end
R3 — Production & Expenses	Batches, items received, expenses	Ingredient consumption and costs tracked

R4 — Reporting & Controls	All report views, exports, audit log, trash, notifications	Owner answers all briefing questions from the app
R5 — Hardening	Offline queue, realtime, storage uploads, PDF export, stock takes, purchase orders	NFR targets met

10. Risks & Mitigations

Risk	Impact	Mitigation
Staff resist data entry	High	Sub-45-second order flow; mobile-first; offline tolerance; training
Poor connectivity	High	Offline queue + retry; local read cache
Stock drift from reality	High	Reason-coded movements; periodic stock take; waste logging
Wrong prices/costs	Medium	Recipe costing; price history; margin report
Unauthorised edits	High	RLS + audit log + owner-only catalog writes
Key person dependency	Medium	Documented setup; exports; no bespoke tooling
Scope creep	Medium	MoSCoW priorities; phased releases

11. Assumptions & Dependencies

Supabase project is available and stays on a paid/adequate tier for backups.

Users have a smartphone or computer with a modern browser and intermittent internet.

The owner will enter the initial product and ingredient catalogue during guided setup.

Naira (₦) is the only currency in v1.

12. Open Questions

Should staff be able to complete orders, or only owners? (Affects M6-FR-08.)

Is VAT registration in place, or should tax default to 0%? (Affects M11-FR-01, M12.)

Are deposits mandatory for custom cakes above a value threshold? (Affects M7-FR-01.)

Which states/cities should be pre-loaded for location reporting?

Should quotes be convertible to invoices in one click? (Recommended: yes.)

Is a printed thermal receipt (58/80 mm) required, or is A4/PDF sufficient?

13. Glossary

Term	Meaning
Ready-made	A cake baked in advance and held in finished-goods stock.
Custom	A made-to-order cake; no stock held; free-form brief.
BOM / Recipe	Bill of materials — ingredients and quantities per unit of product.
COGS	Cost of goods sold — ingredient cost of what was sold.
Stock movement	A single logged change to a stock quantity, with a reason.
Stock take	Physical count compared with system quantity to find variance.
RLS	Row Level Security — Postgres rules that filter rows per user.
Soft delete	Marking a row deleted (deleted_at) instead of removing it.